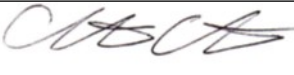


Guideline Title: Pediatric Infectious Respiratory Distress

Guideline Number: GUID-3203-PED010

Issue date: 09/09/2014	Replaces Dept. Policy:
Revision dates: 4/30/24	Developed by: Air Care Team
Approved by: Dr. Christopher Hunter, MD Air Care Team Medical Director	Approved by:
Signature: 	Signature:
Department Numbers: 3203	

I. **PURPOSE:** To maintain a patent airway, optimize ventilation/oxygenation, decrease the work of breathing, treat appropriate respiratory diseases with antibiotics, antipyretics when indicated, and to maintain adequate hemodynamic parameters for the patient

II. **PATHOPHYSIOLOGY:**

Croup is a viral infection that causes erythema and edema of the vocal cords & subglottic region. It typically affects children 3 months to 3 years, and is characterized by a harsh, barking cough, frequently the child is not toxic appearing.

Epiglottitis is a bacterial infection of the epiglottis and larynx. It usually affects children 3 to 7 years of age. Common presentation: toxic appearance, sudden onset with a high fever, muffled voice, and drooling.

Bronchiolitis causes mucous plugging and inflammation of the bronchi leading to hypoxia and atelectasis. Respiratory syncytial virus (RSV) common cause. Most often seen in <2 years of age. There is a history of gradual onset of cough, expiratory wheezing, & increased work of breathing with nasal flaring & retractions.

Pneumonia may be viral or bacterial. Influenza virus most often causes viral pneumonia with an acute inflammatory response which develops in the lungs. Bacterial pneumonia in a non-septic child is usually lobar and may cause a hypoxia. Decreased pulmonary compliance can occur with resultant respiratory distress/failure.

III. **DEFINITIONS:**

IV. **DEPARTMENT GUIDELINE:**

- A. Assess and manage airway, breathing, and circulation. Assess CNS by consciousness and muscle tone
- B. Position child to maximize airway opening. Allow child to assume position of greatest comfort if possible and appropriate.
- C. Supplemental oxygen should be given. Monitor oxygen saturation and end tidal CO₂.
- D. Airway should be secured with intubation, if necessary, *per GUID3203-G008 Rapid Sequence Induction for Endotracheal Intubation and GUID3203-PR020 Endotracheal Intubation*. In the case of epiglottitis, if there is an available anesthesiologist or otorhinolaryngologist, it is recommended the patient be evaluated to determine whether the patient requires immediate airway management in the OR prior to transport. Always have an ETT 1/2 mm and 1 mm smaller than usual ready in case of edematous anatomy.

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- E. For total obstruction, a needle or surgical cricothyrotomy may need to be performed *per GUID3203-PR007 Cricothyrotomy – Surgical*.
- F. Most optimal therapy for epiglottitis includes maintaining a calm environment, avoiding invasive events, and airway evaluated by most experienced person available. Intubate if epiglottitis is present. Confirm ETT with CXR and respiratory exam.
- G. Administer antibiotics per order of referring or receiving physician.
- H. Stridor should be treated with **Racemic Epinephrine 2.25% 0.05 mL/kg** to a maximum single dose of 0.5 ml diluted in 3ml of 0.9% NS and nebulized. May repeat in 20 minutes once. See the *GUID3203-PR012 - Inhalation Therapy*.
- I. Children with wheezing should be treated with bronchodilators per *GUID3203-PED014 - Pediatric Patient Reactive Airway Disease*.
- J. Consider steroids for croup only, **Dexamethasone 0.6 mg/kg IV/oral/IM** to maximum single dose of 10mg.
- K. Use caution to reduce agitation (during attempts for intravenous access, etc.) as this can cause airway decompensation
- L. Treat for shock as outlined in the *GUID3203-PED009 – Hypotension*.
- M. If any question of a bacterial process, antibiotic therapy should be initiated on interfacility transports prior to transport and after blood cultures obtained per *GUID3203-PED016 - Pediatric Sepsis/Meningitis*.
- N. Antipyretics may be given for febrile children prior to transport on inter-facility transports. (**Acetaminophen 15 mg/kg or Ibuprofen 10 mg/kg PO**). Consider rectal administration of acetaminophen if child is having difficulty swallowing or severe stridor.
- O. For inter-facility transport, assess arterial blood gas and chest X-ray.
- P. Follow Orlando Health best practices based for contact, droplet, or airborne precautions based on suspected organism.

V. DOCUMENTATION:

- A. EMS Charts

VI. REFERENCES:

- A. Pediatric Advanced Life Support Manual, 2020. American Heart Association.